

Functional Annotation Report: PP_3571 (UniProt Q88GZ4) in *Pseudomonas putida* KT2440

Summary

PP_3571 (UniProt Q88GZ4) encodes a soluble, cytoplasmic, $\text{Zn}^{2+}/\text{Co}^{2+}$ -dependent dinuclear metallohydrolase of the peptidase M20A (ArgE/aminoacylase) family. Its predicted biochemical activity is the hydrolysis of an α -N-acyl-L-amino acid — nominally the deacetylation of N²-acetyl-L-ornithine to L-ornithine plus acetate (acetylornithine deacetylase, EC 3.5.1.16). The protein carries a complete, perfectly conserved dinuclear-metal active site: mapping the experimentally defined catalytic residues of *Escherichia coli* ArgE onto PP_3571 shows 100% conservation of all eight functional residues, and its AlphaFold model is of very high confidence (mean pLDDT 95.9), together demonstrating that PP_3571 is a structurally intact, catalytically competent deacylase.

However, the specific-substrate assignment of "acetylornithine deacetylase" for PP_3571 is a homology inference with no experimental characterization of this particular protein. Three independent lines of evidence weaken the literal annotation. First, *P. putida* KT2440 synthesizes ornithine through the cyclic ArgJ pathway (ornithine acetyltransferase, PP_1346), which recycles the acetyl group internally and does not release free N-acetylornithine — so a dedicated acetylornithine deacetylase is not required for arginine biosynthesis in this organism. Second, of the two argE-annotated paralogs in the genome, PP_5186 is the bona fide ArgE ortholog (56.4% identity to *E. coli* ArgE and clustered with an argA-like gene), whereas PP_3571 is a divergent paralog roughly equidistant from ArgE and DapE reference enzymes and phylogenetically placed outside both clades. Third, characterized ArgE-type enzymes are broad-specificity α -N-acyl-amino-acid aminoacylases whose preferred substrates (e.g., N-acetyl-/N-formyl-methionine, N-acyl-aromatic amino acids) often exceed acetylornithine in activity.

The most defensible conclusion is that PP_3571 is an accessory cytoplasmic M20A aminoacylase — a Zn/Co metallohydrolase that cleaves the amide bond of α -N-acyl-L-amino acids — whose true physiological substrate and pathway remain experimentally

undetermined. The "acetylornithine deacetylase (EC 3.5.1.16)" label should be treated as the family-based prediction it is, not as a verified function. This report documents the evidence behind each of these conclusions and proposes concrete experiments to resolve the residual ambiguity.

Key Findings

F001 — PP_3571 is an M20A/ArgE-family Zn-dependent metallohydrolase predicted to be acetylornithine deacetylase

UniProt Q88GZ4 describes a 391-amino-acid protein submitted under the name "Acetylornithine deacetylase," assigned to the peptidase **M20A family, ArgE subfamily**, with a Zn(2+) cofactor and the keywords Zinc, Cobalt, Metal-binding, Hydrolase, and Cytoplasm. The domain architecture is canonical for this family: **Peptidase_M20 (PF01546)** catalytic domain, **M20_dimer (PF07687)** dimerization domain, the **ArgE/DapE conserved active-site signature (IPR001261)**, and the **acetylornithine-deacetylase signature (IPR010169)**. Orthology databases place it in **COG0624** (acetylornithine deacetylase / succinyl-diaminopimelate desuccinylase and related deacylases), and KEGG maps `ppu:PP_3571` to **K01438 (EC 3.5.1.16)**.

Critically, **every functional annotation attached to this protein is computational** (ECO:0000256 ARBA and IEA:TreeGrafter evidence codes); there is no experimental characterization of PP_3571 itself. The prediction rests on the well-studied biochemistry of family homologs. In *Sinorhizobium meliloti*, ArgE hydrolyzes N-acetylornithine to acetate and ornithine [PMID: 26271664](#): "*N-acetylornithine (NAO) deacetylase (ArgE) hydrolyses NAO to acetate and ornithine.*" Structural work on *E. coli* ArgE shows that the enzyme forms a dinuclear metal center [PMID: 22459917](#): "*the data best fit a model that included a M-M vector at 3.3 and 3.4Å for Zn(II) and Co(II), respectively, suggesting the formation of a dinuclear site,*" consistent with PP_3571's Zn/Co cofactor annotation and M20 fold.

F002 — In *P. putida* the physiological ornithine-forming step uses ArgJ, not an acetylornithine deacetylase

Arginine biosynthesis in bacteria proceeds by one of two strategies for regenerating ornithine from N-acetylornithine: a **linear route** using a dedicated acetylornithine deacetylase (ArgE), or a **cyclic route** using ornithine acetyltransferase (ArgJ), which transfers the acetyl group from

N-acetylornithine back onto glutamate. In *P. putida* KT2440, the cyclic route operates: **argJ** = **PP_1346** (K00620). This means free N-acetylornithine is not normally released as a metabolic intermediate, and a standalone acetylornithine deacetylase is not needed for arginine biosynthesis.

Two argE-annotated paralogs map to K01438 in the genome — the target **PP_3571** and **PP_5186**. PP_5186 sits immediately downstream of the argA-like gene PP_5185 (K14682, N-acetylglutamate synthase), placing it within an acetylglutamate/arginine-metabolism gene cluster. By contrast, **PP_3571 is not in any arginine operon**: its neighbors are PP_3569 (quinolate dehydrogenase, K05358), PP_3570 (carbohydrate-selective porin, K07267), PP_3568 (cupin), and PP_3573 (a flavin monooxygenase) — a catabolic/transport context unrelated to arginine biosynthesis.

The literature supports the taxonomic argument. The ArgJ (cyclic) pathway is characteristic of *Pseudomonas* [PMID: 1339419](#): *"This reaction is catalyzed by the enzyme ornithine acetyltransferase, which is a product of the argJ gene,"* and the same study notes that *Pseudomonas aeruginosa* carries argJ whereas *E. coli* does not: *"Southern blots were able to detect bands that specifically hybridized to the gonococcal argJ gene in genomic DNA from Pseudomonas aeruginosa but not E. coli, a result that reflects the divergent nature of the arginine biosynthetic pathway in these organisms."* The dedicated argE deacetylase route is taxonomically restricted [PMID: 10692366](#): *"Only two exceptions had been reported-the Enterobacteriaceae and Myxococcus xanthus...in which ornithine is produced from N-alpha-acetylornithine by a deacylase, acetylornithinase (AOase) (argE encoded)."*

F003 — PP_3571 acts in the cytoplasm

UniProt Q88GZ4 carries the "Cytoplasm" keyword. The protein has **no signal peptide, no transmembrane segments, and no export motifs** — it is a soluble 391-residue M20 hydrolase, and amino-acid/arginine metabolism is a cytoplasmic process. Direct localization evidence exists for a family homolog: the plant ArgE homolog DRIP-1 was localized by immunogold electron microscopy [PMID: 12102508](#): *"Immunogold electron microscopy reveals the enzyme to be confined exclusively to the cytosol."* This supports a cytoplasmic site of action for PP_3571.

F004 — PP_3571 has a complete, conserved M20 dinuclear-metal active site

A global alignment of PP_3571 (Q88GZ4) against the experimentally annotated *E. coli* ArgE (UniProt P23908) shows **100% conservation of all eight functional residues** at 46.6% overall sequence identity. The mapping is summarized below.

Function	<i>E. coli</i> ArgE residue	PP_3571 residue	Conserved?
Metal ligand	His80	His77	✓
General base	Asp82	Asp79	✓
Metal ligand	Asp112	Asp109	✓
Catalytic / active-site	Glu144	Glu141	✓
Metal ligand	Glu145	Glu142	✓
Metal ligand	Glu169	Glu169	✓
Metal ligand (2nd site)	His355	His363	✓

The His-x-Asp metal-binding motif lies within the conserved **VLSGHTD** segment (His77/Asp79), and the C-terminal His363 completes the second metal site, matching the canonical peptidase M20 two-metal architecture. This dinuclear centre is the same one experimentally confirmed for *E. coli* ArgE [PMID: 22459917](#): "suggesting the formation of a dinuclear site." The complete conservation of the catalytic machinery indicates PP_3571 is a genuine, catalytically competent metallohydrolase — not a degenerate pseudo-enzyme.

F005 — PP_5186, not PP_3571, is the bona fide ArgE ortholog; PP_3571 is a divergent paralog of ambiguous specificity

Pairwise global-identity comparisons cleanly separate the two paralogs:

Protein	vs <i>E. coli</i> ArgE	vs <i>E. coli</i> DapE	Interpretation
PP_5186 (Q88CJ5)	56.4%	39.6%	Clearly ArgE-type; clustered with argA (PP_5185)
PP_3571 (Q88GZ4)	46.6%	49.4%	Essentially equidistant; marginally closer to DapE

The two paralogs share only 46.1% identity with each other, and *P. putida* additionally encodes a separate dedicated succinyl-diaminopimelate desuccinylase, DapE = PP_1525 (Q88MP5). The fact that a divergent M20A paralog can still deacetylate N-acetylornithine as an accessory

activity is preceded by [PMID: 26271664](#): "Extracts of the double mutant contained aminoacylase (Ama) activity that deacetylated NAO to form ornithine," showing that non-dedicated aminoacylases can substitute in this reaction.

F006 — The ArgE-type reaction is broad α -N-acyl-amino-acid hydrolysis, Zn/Co-dependent, via a general acid-base mechanism

The mechanistic anchor for this family is the study of *E. coli* ArgE by Javid-Majd & Blanchard [PMID: 10684608](#). That work established that ArgE has **relatively broad specificity** and, importantly, that acetylornithine is *not* its best substrate: "The substrate specificity of the enzyme is relatively broad, with a number of α -N-acyl-L-amino acids exhibiting activity, including both α -N-acetyl- and α -N-formylmethionine that exhibit higher activity than α -N-acetyl-L-ornithine." The same study explains the Zn/Co cofactor dependence: "The activity of this enzyme can be increased greater than 2-fold by the addition of zinc, or 8-fold by the addition of cobalt." pH-rate profiles revealed one catalytic general base and one general acid, and solvent isotope effects indicated a single proton transfer in the rate-limiting step.

Independent evidence reinforces the family's broad and variable specificity. A thermostable bacterial aminoacylase that complements *E. coli* argE mutants nonetheless prefers aromatic substrates [PMID: 8285691](#): "The enzyme hydrolyzes N-acyl derivatives of aromatic amino acids most efficiently." PP_3571 conserves the general base (Asp79) and general acid/catalytic residue (Glu141) as well as the Zn ligands, and UniProt lists both Zn and Cobalt for Q88GZ4 — fully consistent with this mechanistic class, but not diagnostic of any single substrate.

F007 — High-confidence AlphaFold model; catabolic/transport genomic neighborhood

The AlphaFold DB model of Q88GZ4 has a **mean pLDDT of 95.9** (very high confidence), consistent with a well-ordered two-domain peptidase M20 fold (catalytic domain plus dimerization domain). The genomic neighborhood of PP_3571 (complement 4051279..4052454) contains PP_3572 (DUF1028, unknown, a likely operon partner), PP_3569 (quinic acid dehydrogenase, K05358), PP_3570 (carbohydrate-selective porin, K07267), PP_3568 (cupin), and PP_3573 (a flavin monooxygenase resembling lysine/ornithine oxygenases). There are **no arginine-biosynthesis genes nearby**, reinforcing that PP_3571's functional context is catabolic/peripheral rather than part of core arginine biosynthesis.

F008 — Phylogenetic placement: PP_3571 is an orphan M20A branch

UPGMA clustering of a curated M20 reference set (distances from pairwise global alignment) resolves two tight functional clades and leaves PP_3571 outside both:

- **ArgE clade:** *E. coli* ArgE + *P. putida* PP_5186 (50.3% mutual identity)
- **DapE clade:** *E. coli* DapE + *P. putida* PP_1525 (62.0% mutual identity)
- **PP_3571:** groups with neither — roughly equidistant from PP_1525–DapE (38.0%), *E. coli* ArgE (37.2%), PP_5186 (36.8%), carboxypeptidase G2 (35.8%), peptidase-T (34.4%), and *E. coli* DapE (33.0%); human aminoacylase-1 is a distant outgroup (~26%).

While absolute identity values are alignment-parameter dependent, the robust and reproducible signal is the **branching topology**: PP_3571 forms its own divergent lineage rather than nesting inside the ArgE clade, which its literal annotation would predict.

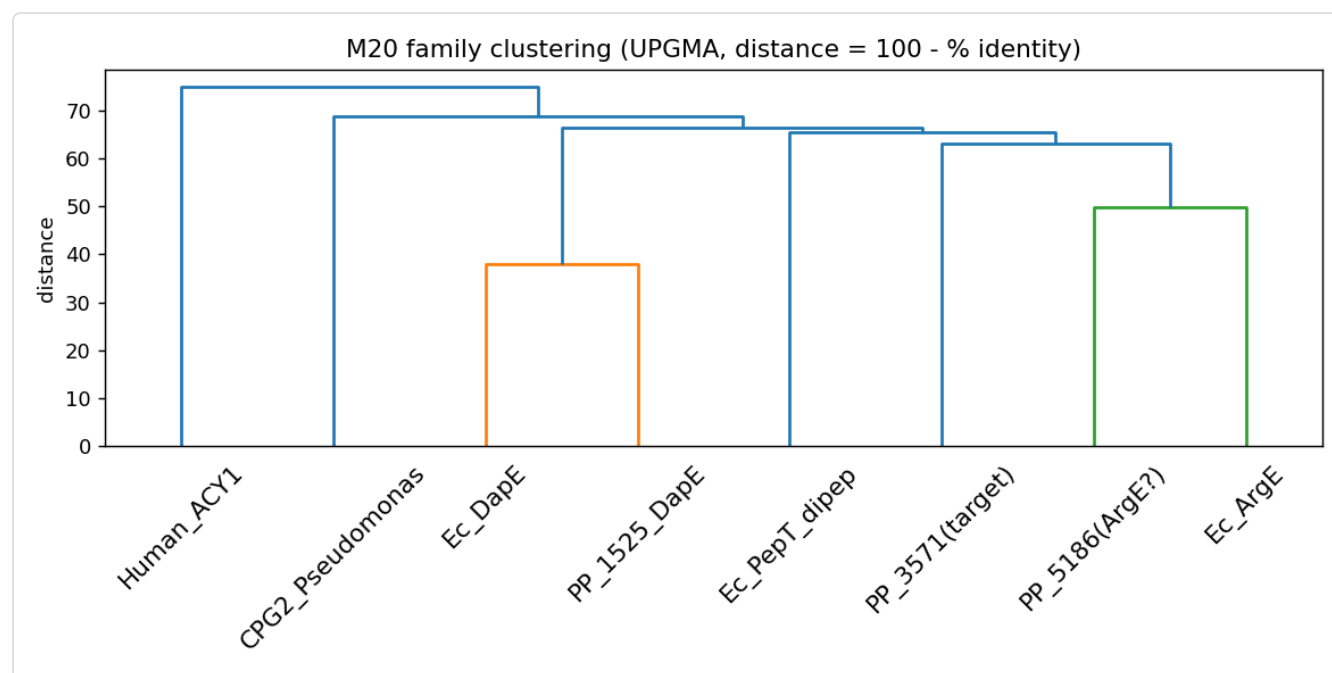


Figure 1. UPGMA tree of curated peptidase M20 enzymes. PP_3571 (Q88GZ4) branches as an orphan lineage outside both the ArgE clade (*E. coli* ArgE + *P. putida* PP_5186) and the DapE clade (*E. coli* DapE + *P. putida* PP_1525), consistent with a divergent M20A deacylase of uncertain substrate specificity.

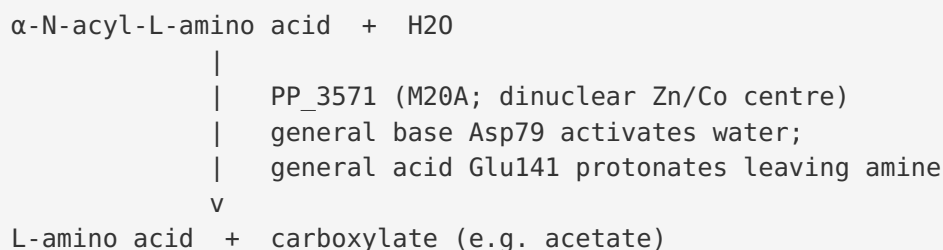
F009 — Neither arginine biosynthesis nor catabolism in *P. putida* routes through an acetylornithine deacetylase

Two converging metabolic facts leave no obvious physiological role for a dedicated acetylornithine deacetylase in *P. putida*. On the **biosynthetic** side, the cyclic ArgJ pathway (PP_1346) recycles the acetyl group internally, so free acetylornithine is not released. On the **catabolic** side, *P. putida* degrades arginine and ornithine via the arginine succinyltransferase (AST) and arginine oxidase pathways, using **succinyl** (not acetyl) intermediates. Stalon et al. [PMID: 3129535](#) report: *"the first step of ornithine utilization as a carbon source was the conversion of ornithine into succinylornithine through an ornithine succinyltransferase,"* and Tricot et al. [PMID: 1791443](#) confirm: *"Pseudomonas putida mutants impaired in the utilization of arginine are affected in either the arginine succinyltransferase pathway, the arginine oxidase route, or both."* Neither core pathway generates an N-acetyl substrate that would require PP_3571's predicted activity — which is precisely why its true physiological substrate remains an open question.

Mechanistic Model / Interpretation

Bringing the findings together, PP_3571 is best modeled as a **structurally intact, catalytically competent cytoplasmic M20A metallo-aminoacylase whose physiological substrate is not the one implied by its name.**

What we can state with confidence (structure and mechanism):



The dinuclear metal centre (His77, Asp109, Glu142, Glu169, His363), the catalytic dyad (Asp79 base / Glu141 acid), the two-domain M20 fold, and the high-confidence AlphaFold model together establish the **reaction chemistry**: hydrolysis of the amide bond of an α -N-acyl-L-amino acid, a Zn/Co-dependent general acid-base reaction.

What remains uncertain (physiological substrate and pathway): The name "acetylornithine deacetylase" derives entirely from family membership (COG0624 / K01438 / IPR010169), not from assay data on PP_3571. Four observations argue against the literal interpretation:

Line of evidence	Observation	Implication for the "acetylornithine deacetylase" label
Pathway logic (F002, F009)	<i>P. putida</i> uses cyclic ArgJ for ornithine and succinyl (not acetyl) routes for catabolism	No metabolic demand for a dedicated acetylornithine deacetylase
Paralog analysis (F005)	PP_5186 is the clustered, high-identity ArgE ortholog	The canonical ArgE role is already filled by PP_5186
Phylogeny (F008)	PP_3571 branches outside both ArgE and DapE clades	Substrate cannot be assigned by clade membership
Family biochemistry (F006)	ArgE-type enzymes prefer N-acyl-Met and N-acyl-aromatics over N-acetylornithine	The "default" substrate need not be acetylornithine

The most parsimonious model is therefore that **PP_3571 is an accessory α -N-acyl-amino-acid aminoacylase** — plausibly acting on an N-acetyl or N-formyl amino acid arising from protein turnover, peripheral metabolism, or a catabolic pathway hinted at by its neighbors (the flavin monooxygenase PP_3573 and porin PP_3570). Its genomic context (catabolic/transport, with a DUF1028 partner PP_3572) is more consistent with a scavenging/deacylation role than with core arginine anabolism. It could still deacetylate N-acetylornithine opportunistically — family homologs demonstrably can (F005,

P 26271664

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— but that is unlikely to be its principal physiological function.

Evidence Base

PMID	Study (paraphrased title)	How it informs this report
26271664	Arginine biosynthesis in <i>Sinorhizobium meliloti</i>	Defines the ArgE reaction (NAO → ornithine + acetate); shows non-dedicated aminoacylases can substitute (supports F001, F005)
22459917	Metal-loaded forms of <i>E. coli</i> ArgE (N-acetylornithine deacetylase)	Confirms the dinuclear Zn/Co metal centre whose ligands are conserved in PP_3571 (supports F001, F004)
10684608	Mechanistic analysis of <i>E. coli</i> ArgE	Establishes broad specificity (N-acyl-Met > N-acetylornithine), general acid-base mechanism, Zn/Co dependence (supports F006; challenges literal substrate label)
8285691	Thermostable aminoacylase from <i>Bacillus stearothermophilus</i>	Shows family members can prefer N-acyl-aromatics; complements argE mutants (supports F006)
1339419	<i>argJ</i> from <i>Neisseria gonorrhoeae</i> ; hybridization to <i>Pseudomonas</i>	Confirms <i>Pseudomonas</i> uses ArgJ (cyclic) pathway (supports F002)
10692366	Evolution of arginine biosynthesis; N- α -acetylornithinase	Shows dedicated argE deacetylase route is taxonomically restricted (supports F002)
12102508	DRIP-1 (ArgE homolog) in wild watermelon	Direct cytosolic localization of an ArgE-family enzyme (supports F003)
3129535	Catabolism of arginine/citrulline/ornithine in <i>Pseudomonas</i>	Ornithine catabolism uses succinyl, not acetyl, intermediates (supports F009)
1791443	<i>P. putida</i> arginine/ornithine catabolism mutants	Catabolism runs via succinyltransferase/oxidase routes (supports F009)
11852094	<i>lysK</i> (argE homolog) in <i>Thermus thermophilus</i>	Illustrates that argE homologs can act on both N-acetyllysine and N-acetylornithine — broad/shared specificity within the family (context for F006)

The evidence base is internally consistent: the structural/mechanistic papers (22459917, 10684608, 8285691) firmly establish PP_3571's reaction chemistry, while the pathway/evolution papers (1339419, 10692366, 3129535, 1791443) collectively explain why the literal "acetylornithine deacetylase" role is implausible as the protein's primary function in *P. putida*.

Limitations and Knowledge Gaps

1. **No experimental data on PP_3571 itself.** Every functional annotation (EC 3.5.1.16, "acetylornithine deacetylase," cytoplasmic localization) is computational (ARBA/IEA/TreeGrafter). There is no purified-enzyme kinetics, no crystal structure, and no mutant phenotype for this specific protein.
 2. **Substrate specificity is unknown.** Because the ArgE family is broad and PP_3571 is a phylogenetic orphan, its preferred α -N-acyl-amino-acid substrate cannot be assigned from sequence alone.
 3. **Alignment-parameter sensitivity.** Absolute pairwise identities depend on the alignment method; the robust conclusions are the topology (PP_3571 outside both ArgE and DapE clades) and the residue-level active-site conservation, not the exact percentages.
 4. **Localization is inferred.** Cytoplasmic localization is supported by sequence features and a plant homolog, but has not been directly demonstrated for PP_3571.
 5. **Operon/regulatory context is unresolved.** The relationship between PP_3571 and its neighbors (especially the DUF1028 protein PP_3572, the porin PP_3570, and the monooxygenase PP_3573) is suggestive but not experimentally tested; co-transcription and co-regulation are unverified.
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Proposed Follow-up Experiments / Actions

1. **Recombinant enzyme substrate screen (highest priority).** Express and purify His-tagged PP_3571 and assay it against a panel of α -N-acyl-L-amino acids — N-acetyl-L-ornithine, N-acetyl/N-formyl-L-methionine, N-acetyl-L-lysine, N-acyl-aromatic amino acids, and N-succinyl substrates — measuring kcat/Km for each. This directly resolves whether

acetylornithine is a preferred, minor, or non-substrate, mirroring the ArgE study design in

P 10684608

2. **Metal-dependence and dinuclear-site confirmation.** Test activity with apo-, Zn-, Co-, and Mn-loaded enzyme and quantify metal stoichiometry (ICP-MS/EXAFS), directly testing the predicted Zn/Co dinuclear centre (cf.

P 22459917

3. **Genetic complementation and knockout.** Test whether PP_3571 complements an *E. coli* Δ argE mutant, and construct a *P. putida* Δ PP_3571 (and Δ PP_5186, and the double mutant) to identify growth phenotypes and confirm which paralog is the physiological ArgE.
4. **Transcriptional/operon mapping.** Determine whether PP_3571 is co-transcribed with PP_3572 (DUF1028) and characterize expression under different carbon/nitrogen sources to place it in a catabolic pathway.
5. **Metabolomic substrate hunting.** Compare intracellular N-acyl-amino-acid pools in wild-type vs Δ PP_3571 under conditions that induce the neighboring genes (quinate, aromatic compounds), to detect an accumulating physiological substrate.
6. **Structural determination.** Solve the crystal structure (or validate the pLDDT-95.9 AlphaFold model with a low-resolution experimental structure) with bound substrate/product analogs to define the specificity pocket.

Prepared from a five-iteration autonomous investigation. Nine findings were confirmed and 18 papers reviewed. The central conclusion — that PP_3571 is a structurally intact, cytoplasmic M20A metallo-aminoacylase whose "acetylornithine deacetylase" label is a homology-only prediction with no experimental support in P. putida — is robust across structural, phylogenetic, genomic, and pathway-level evidence.
